

Ultimate Linux Commands Cheat Sheet

Boost your **Linux skills** with these must-know commands for file management, networking, users, system monitoring, and troubleshooting!



File & Directory Management

- ♦ **ls -lAh** → List all files with details
- ♦ **tree /path/to/dir** → Show directory structure
- ♦ **find /home -name "*.log"** → Find files by name
- ♦ **du -sh /var/log** → Check folder size
- ♦ **rsync -av source/ dest/** → Sync files between directories
- ♦ **stat file.txt** → Show detailed file metadata

File Permissions & Ownership

- ♦ **chmod 644 file** → Set file permissions (Owner: RW, Others: R)
- ♦ **chown user:group file** → Change file ownership
- ♦ **umask 022** → Default permission for new files

User & Group Management

- ♦ **who** → Show logged-in users
- ♦ **last** → Show login history
- ♦ **sudo userdel -r user** → Delete user & home directory
- ♦ **groups user** → List groups of a user

- ♦ **sudo visudo** → Edit sudo privileges

Process Management

- ♦ **ps aux | grep apache** → Find processes by name
- ♦ **top** → Real-time system resource monitoring
- ♦ **htop** → Advanced process viewer (if installed)
- ♦ **vmstat 2 5** → Show CPU & memory stats
- ♦ **iostat -dx 2** → Show disk I/O stats

Package Management

Debian/Ubuntu (apt):

- ✓ **sudo apt search package_name** → Search for a package
- ✓ **sudo apt list --installed** → Show installed packages

♦ CentOS/RHEL (yum/dnf):

- ✓ **sudo yum list installed | grep nginx** → Check if a package is installed
- ✓ **sudo dnf autoremove** → Remove unused packages

♦ Snap (for universal packages):

- ✓ **snap list** → List installed snap packages

Disk & Storage

- ✓ **df -h** → Show disk usage
- ✓ **du -sh folder** → Check folder size
- ✓ **mount /dev/sdb1 /mnt** → Mount a drive
- ✓ **umount /mnt** → Unmount a drive

Networking Commands

- ✓ **ip a** → Show IP address
- ✓ **ping google.com** → Test network connectivity
- ✓ **netstat -tulnp** → Show open ports
- ✓ **scp file user@remote:/path/** → Securely copy files
- ✓ **wget URL** → Download file from a URL

Log & System Monitoring

- ✓ **dmesg | grep error** → Check boot errors
- ✓ **journalctl -u apache2 --since today** → View service logs
- ✓ **tail -f /var/log/syslog** → Live log monitoring

Backup & Restore

- ✓ **rsync -avz /source/ /backup/** → Backup files
- ✓ **tar -czvf backup.tar.gz /data/** → Create a compressed backup
- ✓ **dd if=/dev/sda of=/backup.img** → Create a full disk image
- ✓ **scp user@remote:/path/file .** → Securely copy a file from remote

Automation & Scripting

- ✓ **echo "Hello, World!" > file.txt** → Write to a file
- ✓ **cat file.txt** → Display file contents
- ✓ **sed -i 's/old/new/g' file.txt** → Replace text in a file
- ✓ **awk '{print \$1}' file.txt** → Print first column of a file
- ✓ **crontab -e** → Edit cron jobs
- ✓ **crontab -l** → List scheduled jobs

Compression & Archiving

- ✓ **tar -cvf archive.tar folder/** → Create a tar archive
- ✓ **tar -xvf archive.tar** → Extract a tar archive
- ✓ **zip -r archive.zip folder/** → Create a zip archive
- ✓ **unzip archive.zip** → Extract a zip file

Pro Tip: Use `man <command>` for more details on any Linux command! 